A selected excitation and its curvature dynamics

The complete covariant density in complex double-scaled SYK

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Abstract

We identify the restriction of one Slater-selected excitation to the entire covariant algebra generated by the Hamiltonian, size fluctuation and same-disorder quadratic curvature. Its previously known two-field density remains valid after adding curvature; a boundary-pairing proof establishes this for every fixed mixed word. On the Hamiltonian-curvature subalgebra the state is stationary with density $1-\sqrt{q}+\sqrt{q}\,h^2$. Its curvature marginal is exactly the tracial one, but its time correlation differs. At $q=1/4$ and dimensionless time $t=2$, an exact rational certificate proves that this correlation is negative while the tracial correlation is positive. The preceding polynomial-density estimate connects this kernel to the actual finite-N curvature after exact energy pinching.

1 Microscopic normalization and the density

Use standard CAR, an overlapping Hermitian complex Gaussian p-body Hamiltonian with one common disorder J, a half-filled Slater vector O, and $\varphi_N=\mathbb{E}\langle O|\cdot|O\rangle$. Take $m=N/2$, $p^2/N\to\lambda>0$, at fixed observables and times. Let $C_Q=\mathbb{E}H^2$ and C_m be its charge-m value. The fields are

$$\begin{aligned} h_N &= H_m/\sqrt{C_m}, & W_N &= (p-\mathcal{L}-\lambda_N)H|_m/\sqrt{C_m},\\ j_N &= C_m^{-1}(2p-D-\kappa_N)(H^2-C_Q)|_m, & x_N &= j_N/\sqrt{v_N},\\ DX &= QX-\sum_i c_i^\dagger X c_i, & \mathcal{L}X &= \tfrac12(NX-\sum_i(c_i^\dagger X c_i+c_i X c_i^\dagger)). \end{aligned} \tag{1}$$

Loss maps precede charge restriction; C_m is a fixed denominator. In the limit use

$$q=e^{-4\lambda},\quad \gamma=e^{-\lambda},\quad s=\gamma^2=\sqrt{q},\quad \kappa=2\lambda\frac{1-q}{1+q},\quad v=4\lambda+\frac{16\lambda^2 q}{1+q},\quad W=\sqrt{\lambda}\,u. \tag{2}$$

The subscript-N constants in (1) use (2) at $\lambda_N=p^2/N$. $M^\dagger$ is the normalized occupied-to-empty block of the original J, with $\varphi(MM^\dagger)=1$. Energies and h-times are dimensionless, $\hbar=1$. The prior source theorem identifies the covariant algebra $\mathcal{D}=W^*(h,u,x)$ with a bounded normal representation of the tracial mixed-q law, with crossing entries q,q^2,q^4 for lengths $1,1,2$.

Theorem 1. *For every $B\in\mathcal{D}$, the one-excitation state is*

$$\boxed{\sigma_1(B):=\varphi(MBM^\dagger)=\tau(\rho_1 B),\qquad \rho_1=s(h-W)^2+r_P I,\quad r_P=1-s(1+\lambda)>0.} \tag{3}$$

Here $\tau(\rho_1)=1$ and $\|\rho_1\|=r_P+4s(1+\lambda)/(1-q)$.

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2 Why the two-field density survives the curvature extension

Theorem 1 was previously established on $W^*(h,u)$. We prove the extension directly from the full selected source rule, without extrapolating finite moment tests. Expand each j as an adjacent two-H block with a whole-D source; exclude its internal Gaussian pair. A W is one H slot with a whole-$\mathcal{L}$ source. For a pairing π the earlier formula is

$$\gamma^b[z_{\mathrm{all}}]\exp\left(\lambda K_\pi(z)-\lambda\sum_W z_a-\kappa\sum_j z_a\right), \qquad K_\pi(0)=-4\nu. \tag{4}$$

G denotes an unrestricted edge; F is a forward selected edge. ν counts crossings plus nestings whose inner edge is F. The established geometric source rule gives $K_W=1_G$ and a same-edge WW coefficient 1_G. A quadratic source has coefficient 2ϵ where $\epsilon=+1$ for noncrossing incident edges and -1 for crossing ones, except that same-side partners require a specified incident edge to be G. For right-side partners it is the one with the smaller right endpoint; for left-side partners it is the one with the larger left endpoint. A pair of quadratic sources has coefficient 4 precisely for a complete nested two-edge connection. Mixed Wj and higher source coefficients vanish.

Place the boundary insertions of $MBM^\dagger$ at $0, L$. There are two disjoint cases.

Direct boundary pairing. The edge $(0,L)$ is F and surrounds all other edges, which are G. It has no crossing and is never an inner selected edge. No source touches its endpoints, so this contribution is exactly $\tau(B)$.

Two attached boundary edges. Each boundary now meets an internal endpoint, giving exactly two F edges and a normalization factor $\gamma^2=s$. An edge touching 0 or L cannot be the inner member of any nesting. Thus ν equals the ordinary crossing count, as for the corresponding all-G pairing. Every internal quadratic source also has the all-G coefficient: if an incident F edge has partner 0 and both partners are on the left, the larger left partner belongs to the other, G, edge. The case with partner L and both partners on the right is identical with the smaller right partner. Split partners have no G condition. If both boundary edges meet the same quadratic block, its partners are $0, L$ and are necessarily split. The jj' coefficient is unchanged.

The only altered source is a W connected to a boundary: its linear coefficient drops from 1 to 0, leaving $-\lambda$ after centering. Now compare with the all-G word $\chi B\chi$, $\chi=h-W$. A boundary χ attached to an internal H or quadratic slot has weight 1. Attached to an internal W it has weight $-\lambda$: HW vanishes and the $-WW$ term contributes $-\lambda$. All remaining source factors agree. The directly paired boundaries of $\chi B\chi$ have weight $\tau(\chi^2)=1+\lambda$. Consequently

$$\sigma_1(B)=\tau(B)+s\big[\tau(\chi B\chi)-(1+\lambda)\tau(B)\big]. \tag{5}$$

Trace cyclicity proves (3) for all polynomials. Both sides are normal on the established covariant representation, so the identity extends to $\mathcal{D}$. The q-Gaussian law of χ has variance $1+\lambda$, giving the norm in the theorem. Finally $e^{2\lambda}>1+\lambda$ proves $r_P>0$.

3 Restrictions of the state and observable-dependent stationarity

Let $\mathcal{D}_{hx} = W^*(h, x)$. For a word B in h, x, the mixed-q rule annihilates terms with a single W. The two adjacent W endpoints in W^2B pair without crossing and give $\lambda\tau(B)$. Therefore

$$\boxed{\sigma_1(B) = \tau(f(h)B), \quad B \in \mathcal{D}_{hx}, \qquad f(h) = 1 - s + sh^2.} \tag{6}$$

The extension to bounded B again follows by normality. $f(h) \geq (1-s)I > 0$ and it commutes with h. Hence this restricted state is stationary under $\alpha_t(B) = e^{ith}Be^{-ith}$.

For B containing only x, the two h endpoints in h^2B also pair without crossing. Thus

$$\sigma_1|_{W^*(x)} = \tau|_{W^*(x)}. \tag{7}$$

The *entire* normalized curvature distribution is the unit-variance q^4-Hermite law, supported on $[-2/\sqrt{1-q^4}, 2/\sqrt{1-q^4}]$. This is stronger than agreement of a few equal-time moments. The Hamiltonian marginal, however, is $f(E)\mu_q(\mathrm{d}E)$, so that

$$\sigma_1(h^2) = 1 + s(1+q). \tag{8}$$

The full state on $\mathcal{D}$ is not stationary. A direct invariant diagnostic is

$$\boxed{\tau([h, \rho_1]^\dagger[h, \rho_1]) = 2q\lambda(1+\lambda)(1+q)(1-q^2) > 0.} \tag{9}$$

For a derivation rotate the two unit-variance q-Gaussian fields h, u: write $z = \chi/\sqrt{1+\lambda}$ and choose an orthogonal unit field y such that

$$h = \frac{\chi}{1+\lambda} + \sqrt{\frac{\lambda}{1+\lambda}}\, y. \tag{10}$$

The two-field q-Gaussian law is invariant under this orthogonal rotation. The mixed moments needed for the commutator are

$$\tau(y^2z^4) = 2 + q, \qquad \tau(yz^2yz^2) = 1 + q^2 + q^3. \tag{11}$$

Their difference gives $\|[y, z^2]\|_2^2 = 2(1+q)(1-q^2)$, and the factors $s^2\lambda(1+\lambda)$ give (9). Since τ is faithful, the density fails to commute with h.

This distinguishes stationarity on two specified observable algebras: the h, x restriction is stationary, while observables involving u can detect nonstationarity. It does not assign a de Sitter temperature to this preparation. The polynomial density in (6) is not an asserted Gibbs weight.

Covariant dressing. For a bounded covariant A, the preparation $AM^\dagger\Omega$ has unnormalized density $A\rho_1A^\dagger$ on $\mathcal{D}$. Its normalized state divides by $\tau(A\rho_1A^\dagger)$. This follows directly from (3); it does not identify states made with arbitrary interlaced selected insertions. In particular it does not replace the previously nonzero off-diagonal matrix element $\varphi(j(M^\dagger)^2)$.

4 The actual curvature two-point function

The preceding finite-N result gives the normalized actual curvature

$$a_N = \frac{A_{0,N}}{C_m\sqrt{v_N}} = (1 - \mathcal{P}_{H_N})x_N \tag{12}$$

exactly. Its error relative to x_N tends to zero in every fixed central Schatten norm and, by the polynomial orbit-density estimate, in covariant blocks between fixed selected preparations. With $a_N(t) = e^{ith_N}a_N e^{-ith_N}$, that theorem applies to

$$\lim_{N\to\infty} \varphi_N(M_N a_N(t) a_N M_N^\dagger) = C_1(t) := \tau(f(h)e^{ith}xe^{-ith}x). \tag{13}$$

Exact pinching is performed at finite N before the fixed-time double-scaling limit. No long averaging window is substituted here.

Let μ_q denote the unit-variance Hamiltonian measure and H_n its monic orthogonal polynomials, with norm $[n]_q!$. The established Mehler kernel [1, Sec. 3.2] is

$$K_r(E, E') = \sum_{n\geq 0} \frac{r^n H_n(E)H_n(E')}{[n]_q!}, \qquad r = q^2. \tag{14}$$

Its parameter is the Hamiltonian-curvature crossing weight. The selected density, rather than a change to that known kernel, gives

$$\boxed{C_1(t) = \int f(E)e^{it(E-E')}K_{q^2}(E, E')\mu_q(\mathrm{d}E)\mu_q(\mathrm{d}E').} \tag{15}$$

Write C_0 for the same formula with $f = 1$. The measure in (15) is positive and has total mass one. Simultaneous $(E, E') \mapsto (-E, -E')$ symmetry makes C_1 real and even; positive definiteness in time does not require nonnegative values at every time.

The curvature marginal (7) fixes $C_1(0) = 1$ but not the two-energy measure in (15). Its short-time behavior is

$$\boxed{-C_1''(0) = \sigma_1([h, x]^\dagger[h, x]) = 2(1 - q^2) + s(1 + q)(1 - q^2)^2.} \tag{16}$$

To see this, use the first two conditional Mehler moments,

$$\mathbb{E}[E' \mid E] = rE, \qquad \mathbb{E}[(E')^2 \mid E] = 1 - r^2 + r^2E^2. \tag{17}$$

The conditional second moment of $E - E'$ is $(1 - r)^2E^2 + 1 - r^2$. Averaging against $f\mu_q$ gives (16). The additional term is strictly positive for fixed $0 < q < 1$. These are derivatives of the limiting bounded model; no small-t, small-λ, or large-N limits have been exchanged.

For a convenient computational form let T be the q-Jacobi matrix with off-diagonal entries $\sqrt{[n]_q}$, and $D_r = \mathrm{diag}(r^n)$. Since $T^2|0\rangle = |0\rangle + \sqrt{1 + q}|2\rangle$,

$$C_1(t) = C_0(t) + s\sqrt{1 + q}\,\langle 2|e^{itT}D_r e^{-itT}|0\rangle, \quad C_0(t) = \langle 0|e^{itT}D_r e^{-itT}|0\rangle. \tag{18}$$

5 Physical-curve evaluation and the sign change

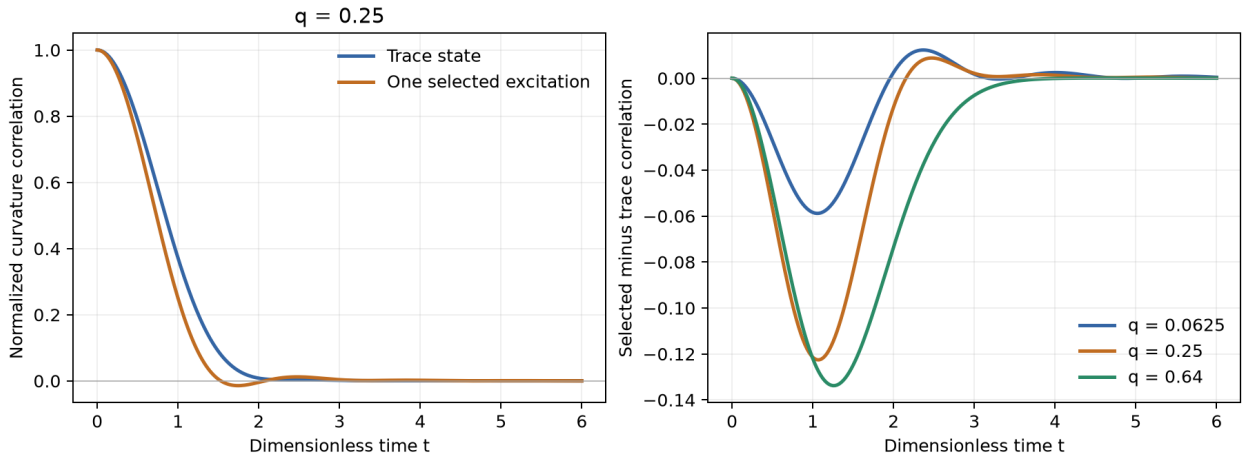


Figure 1. Left: the actual limiting normalized curvature correlation for one selected excitation and for the trace state, at $q = 1/4$. Right: their difference for three physical values of q, with $\lambda = -\log q/4$. Times are conjugate to unit-variance h. The selected correlation crosses below zero; it is a correlation, not a probability density.

q	t	$C_0(t)$	$C_1(t)$
0.0625	1	0.336956823	0.278539693
0.25	1	0.371487074	0.250027382
0.64	1	0.556620677	0.434562095
0.25	2	0.0081500731	−0.0044147484
0.25	4	0.0005801737	0.0019050924

We evaluated (18) at dimensions 192, 256 and independently integrated (15) in q-Hermite angles using Gauss–Legendre orders 192, 288 and product lengths 128, 192. The largest absolute discrepancy between the two representations over the eighteen saved time points was 7.33×10^{-14}. Both comparison methods use floating arithmetic; their agreement is a diagnostic, not an interval certificate.

For clarity the angular measure and kernel used by the independent implementation are the known products

$$E(\theta) = \frac{2\cos\theta}{\sqrt{1-q}}, \qquad \mu_q(\mathrm{d}\theta) = \frac{(q, e^{2i\theta}, e^{-2i\theta}; q)_\infty}{2\pi}\mathrm{d}\theta,$$
$$K_r(\theta,\phi) = \frac{(r^2;q)_\infty}{\prod_{\epsilon,\eta=\pm1}(re^{i(\epsilon\theta+\eta\phi)};q)_\infty}. \tag{19}$$

The integration keeps both positive measure and complex phases. Relative cutoff changes and phase cancellation are recorded separately from absolute errors. The largest saved phase-sum condition number was approximately 4.21×10^4, at $q = 0.64, t = 6$.

The negative value at $q = 1/4, t = 2$ merits a stronger check. The next page gives exact rational bounds for its sign, independently of either floating implementation. This is a distinction between preparations within the specified microscopic limit; a bulk scattering or finite-G interpretation is not required to establish it.

6 A rational sign certificate and explicit truncation bounds

At $q = 1/4$, $s = 1/2$, $r = 1/16$, all polynomial moments of the measure in (15) are rational, even though $\lambda = \log 2/2$. Expand powers in the monic Hermite basis,

$$E^a = \sum_n c_{an} H_n(E), \qquad EH_n = H_{n+1} + [n]_q H_{n-1}. \tag{20}$$

The recurrence computes every c_{an} by rational operations. Orthogonality and Mehler diagonalization give

$$\begin{aligned} B_{ab} &= \int E^a (E')^b K_r \, \mathrm{d}\mu_q \mathrm{d}\mu_q = \sum_n c_{an} c_{bn} [n]_q! r^n, \\ B^{(1)}_{ab} &= (1-s) B_{ab} + s B_{a+2,b}, \\ m^{(i)}_{2k} &= \sum_{a=0}^{2k} (-1)^{2k-a} \binom{2k}{a} B^{(i)}_{a,2k-a}, \qquad B^{(0)} = B. \end{aligned} \tag{21}$$

Thus $m^{(i)}_{2k}$ is the exact moment of $(E - E')^{2k}$ in state $i = 0, 1$. For $t = 2$ and $K = 20$, compute

$$S_i = \sum_{k=0}^{20} \frac{(-1)^k 2^{2k}}{(2k)!} \, m^{(i)}_{2k}. \tag{22}$$

The energy support implies $|E - E'| \le 8/\sqrt{3} < 5$. The degree-41 Taylor formula for cosine (whose odd coefficient is zero), integrated against the positive unit-mass measure, proves

$$|C_i(2) - S_i| \le \frac{10^{42}}{42!} < 7.118 \times 10^{-10}. \tag{23}$$

All operations and interval comparisons use exact fractions. Outward decimal enclosures of the saved rational endpoints are

$$\begin{aligned} 0.0081500723940 < C_0(2) < 0.0081500738176, \\ -0.0044147491323 < C_1(2) < -0.0044147477086. \end{aligned} \tag{24}$$

The sign distinction is therefore rigorous within the established limiting model. No eigensolver, quadrature, or decimal rounding enters the rational sign test. The negative interval also implies eventual negativity of the real part of the finite-N expectation in (13), by its fixed-time convergence.

Jacobi truncation. For a cut at dimension L, powers acting on $|0\rangle$ agree with the infinite matrix through degree $L - 1$; for $|2\rangle$ they agree through degree $L - 3$. Put $z = 2|t|/\sqrt{1-q}$. Taylor tails and $\|D_r\| = 1$ give the valid common bound

$$|C_1(t) - C_{1,L}(t)| \le 4(1 + s\sqrt{1+q}) e^z \frac{z^{L-2}}{(L-2)!}, \qquad z^2 \le L(L-1). \tag{25}$$

Every saved setting satisfies the displayed condition. This is an exact-arithmetic cutoff bound; floating roundoff is separate.

For the product-integral implementation, length K gives the uniform bound on the logarithm of the omitted product factor

$$\eta_K \le \frac{6q^{K+1}}{(1-q)(1-q^{K+1})} + \frac{(r^2 + 4r) q^K}{(1-q)(1-rq^K)}. \tag{26}$$

It follows from the convergent logarithmic series for each Pochhammer factor. Gauss–Legendre discretization is checked by refinement, not claimed to be covered by this product-tail bound.

7 Evidence, provenance and the next physical interface

The source comparison has two levels: an exact boundary-diagram check of the proof's local statement, and an independent comparison of the complete selected transfer with the covariant trace formula.

Exact check	Count
All words of length 0 through 6 in H, W, J	1093
Density comparisons at two independent rational points	2186
Boundary-pairing full source exponents	2953
Reduced H, J density comparisons through length 8	1022
Pure curvature marginal checks through power 12	26
Curvature commutator and full-density commutator formulas	4

The rational pairs are $(\gamma, \lambda) = (1/2, 1/3), (2/3, 1/4)$; they are algebraic tests off the physical curve. Physical positivity and the all-word statement follow from the proof, not from those samples. The numerical dynamics separately uses $q = 0.0625, 0.25, 0.64$ with $\lambda = -\log q/4$.

All three scientific drivers completed successfully on their first full run and their outputs are preserved: exact moment/source checks, physical-curve dynamics and figure, and the rational sign certificate. Input hashes, parameters, dependency versions, cutoff changes, phase-sum conditions, and authoritative rational endpoints are stored alongside the results. Earlier successful scientific runs were not repeated. The accompanying manifest binds this manuscript, the PDF, build records and the source evidence.

The literature supplies the chord/GNS framework [2] and the Mehler spectral kernel [1]. The project-specific result here is the boundary-source identity for the same-disorder composite and the resulting selected actual-curvature response. The old h, u density is used as a predecessor, while its extension to x is proved anew.

The useful next interface is to higher charged and neutral bilocal descendants in this explicit state. The existing full-trace finite-time bilocal construction remains available. One must identify the actual fixed-sector rank/charge maps and the selected-state time tails before claiming that complete observable in the new preparation. Preparations with more selected insertions need their own boundary-source analysis; the two-extreme-endpoint argument above does not automatically cover them. Spatial and mass generators and finite-G dynamics remain the long-term targets of the de Sitter program.

References

[1] M. Berkooz, M. Isachenkov, V. Narovlansky and G. Torrents, *Towards a full solution of the large N double-scaled SYK model*, arXiv:1811.02584v2, Secs. 2.3 and 3.2, especially Eqs. (3.15)–(3.16).

[2] H. Rajgadia and J. Xu, *Emergent States and Algebras from the Double-Scaling limit of Pure States in SYK*, arXiv:2604.14387v3, Sec. 4.